BT 402

M.C.Q file for Midterm

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i. Microorganisms came into being on earth over a period of about **1.2 to 1.5** billion years

ii. Fossil microbes have been found in rocks **3.3 to 3.5** billion years old.
iii. **Rhizobacteria** found in the soil fixates **nitrogen** which is required to grow many crops.
iv. **Microorganisms** are responsible for recycling the huge masses of organic matter synthesized by plants as life on earth evolved.
v. **Cyanobacteria** or blue green algae produce **oxygen** in the ocean.
vi. _____ is the breakdown of raw organic materials to finished compost. **Decomposition**
vii. Without this recycling of inorganic nutrients, primary productivity on the **globe** would stop.
viii. The molecular characterization of all genes in a species is called **Genomic**
ix. GMO stand for _____ **Genetically modified organism**
x. The use of molecular characterization to provide more accurate and quicker identification of **pathogens**
xi. In vaccine technology _____ is used to develop the immunity. **rDNA**
xii. Production of antibiotics, organic acids and enzymes by fermentation of **microbes**
xiii. In agriculture industry the microorganisms are used as **Biofertilizer and Biopecticide**
iv. Ability of microorganisms to degrade toxic materials like **oil, petroleum, plastic**
15. The 1st step of issue in Bioprocessing is **Isolation of Pure or Mixed Cultures from nature**
16. Which one of the following are the physical factors? **Temperature Humidity Ph viscosity**
17. Nutritional requirements can be determined from chemical composition of **microbial cells**.
18. Genetically modified organism (GMO), organism whose genome has been engineered in the laboratory in order to favors the expression of desired **physiological traits** or the production of desired biological products.
19. Which of the following are involved in Genetic modification? **Mutation, insertion and deletion**
20. Inserted genes usually come from a different species in a form of **horizontal** gene transfer.
21. Inserted genes usually come from a same species in a form of **vertical** gene transfer.
22. Introducing DNA from one organism into the cell of another by use of an electric pulse. ***E*lectroporation**
23. **Bacteria** were the first organisms to be modified in the laboratory, due to their simple genetics.
24. Genetically modified bacteria are used to produce the protein insulin to treat **diabetes**.
25. The most common example of GMO's plants are; **Foods – canola , soya bean ,golden rice Sugar beets ,alfalfa**
26. Gene technology is the best solution to the problem of **world hunger**.
27. Genetically engineered animals may suffer more health problems? **Allergic reactions "Genetic Pollution"**
28. Microorganisms are essential for the **earth** to function.
29. **Diversity patterns** of microorganisms can be used for monitoring and predicting environmental change.
30. Microorganisms play a role in **conservation and restoration** biology of higher organisms.
31. Many bacteria help decompose dead organism and animal wastes into chemical compound? **Ammonia**
32. Which one of the following is the main component of plant cell wall? **Cellulose**
33. Which enzyme can digest the cellulose? **Cellulase**
34. Some type of microorganism produce important drugs called **ANTIBIOTICS**
35. Penicillum notatumis is commonly known as **GREEN MOULD**
36. Bacteria break down dead plants and dead animals by releasing **AMMONIA**

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37. The nitrogen fixing bacteria in the soil change the **ammonia into nitrates**
38. Pea and bean plants use bacteria **Rhizobium** which directly convert nitrogen into acid amino.
39. Which gas causes bubbles released in the dough causes it to rise so the bread becomes fluffy? **CO_2**
40. The most important bacteria in food manufacturing are **Lactobacillus** species, also referred to as lactic bacteria.
41. The production of cheese utilizes starter cultures being added to milk with the aim of producing **semi-hard** and hard cheeses.
42. The starter cultures produce lactic acid, resulting in a **pH decrease** in the milk and promoting curd synthesis.
43. Lactic acid generating bacteria cultures **Lactobacillus** and their use of lactose in milk as a substrate to produce lactic acid.
44. **Salami** is a type of cured sausage consisting of fermented and air-dried meat, typically **beef or pork**
45. Ham is pork that has been preserved through **salting, smoking, or wet curing**
46. Yeasts are responsible for the fermentation process which produces **alcohol** in wine.
47. Fermentation is a metabolic process that consumes sugar in the absence of **oxygen**
48. Fermentation sometimes refers specifically to the chemical conversion of **sugar into ethanol**
49. How many types of fermentation? **Two**
50. Which fungi commonly called Brewer's yeast? **Yeast (*Sacharomyces cerevisiae)***
51. **Vitamins** are some organic compounds which are capable of performing many life-sustaining functions inside our body.
52. Which one of the following are the name of Vitamin B_1 and Vitamin B_2? **Thiamine and Riboflavin**
53. Organic solvents such as ethanol, acetone, butanol and glycerolare some very important chemicals that are widely used in **petrochemical industries**
54. ____ are used as **anti-inflammatory drugs**, and as hormones such as **estrogens and progesterone**, which are used in oral contraceptives. **Steroids**
55. **Recombinant DNA technology** is increasingly important in the development and production of subunit vaccines.
56. ____ is used as an immunosuppressive agent in organ-transplant patients. **Cyclosporin A**
57. _____ is acts by competitively inhibiting the enzyme responsible for the synthesis of **cholesterol. Statins**
58. A **vaccine** is a biological preparation that improves immunity to a particular disease.
59. The term of vaccination was 1st devised by? **Edward Jenner**
60. During the breakdown of Glucose into how many pyruvic acids? **Two**

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